

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
		Registration using Gmail
FR-2	User Authentication	Login using Email and Password
		Logout from Application
FR-3	Product Management	View Products
		Search Products
		Filter Products by Category
		View Product Details
FR-4	Cart Management	Add Product to Cart
		Update Product Quantity
		Remove Product from Cart
FR-5	Order Management	Place Order
		View Order History
		Track Order Status
FR-6	User Profile	View Profile Information
		Update Profile Details
FR-7	Admin Authentication	Admin Login
FR-8	Product Administration	Add Product
		Update Product
		Delete Product

		Manage Product Catalog
FR-9	Order Administration	View Customer Orders
		Update Order Status
FR-10	User Administration	View Registered Users
		Manage User Accounts

Non-Functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide an intuitive and user-friendly interface for customers and administrators.
NFR-2	Security	User passwords must be encrypted, and authentication must be implemented using secure JWT-based authorization.
NFR-3	Reliability	The system should consistently perform intended operations without data loss or failures.
NFR-4	Performance	Product searches, page loads, and API responses should be completed within acceptable response times.
NFR-5	Availability	The application should be accessible to users whenever required with minimal downtime.
NFR-6	Scalability	The system should support increasing numbers of users, products, and orders without significant performance degradation.
NFR-7	Maintainability	The codebase should be modular and easy to update, test, and maintain.
NFR-8	Compatibility	The application should work across modern web browsers and devices.
NFR-9	Data Integrity	Product, cart, and order data must remain accurate and consistent throughout transactions.
NFR-10	Backup & Recovery	Database backups should be maintained to prevent data loss and enable recovery.